TP1 Test Cases

Team 4 CSE 360 TP1 Covers: New Username Recognizer, Password Recognizer, Email Address Validator, application-level field validation, and the Phase 1 functional user stories (US-01 through US-08)

1. Username Recognizer

State machine summary: Start → State 1 requires an ALChar (A–Z, a–z). From State 1, UNChar (A–Z, a–z, 0–9) loops back to State 1. A special character (-, _, ., &) moves State 1 → State 2, and State 2 requires a UNChar to return to State 1. A valid username must end in State 1 (i.e., never end on a special character, and never have two special characters in a row).

ID	Input	Expected Result	Rule Being Tested
TC-U01	abcd	Valid	Letters only
TC-U02	abc123	Valid	Letters followed by digits
TC-U03	a1b2c3	Valid	Mixed alphanumeric
TC-U04	user-name	Valid	Single dash between two UNChars
TC-U05	user_name	Valid	Single underscore between two UNChars
TC-U06	user.name	Valid	Single period between two UNChars
TC-U07	user&name	Valid	Single ampersand between two UNChars
TC-U08	a	Invalid	Minimal one-letter username
TC-U09	1username	Invalid	Must start with ALChar, not a digit
TC-U10	-username	Invalid	Must start with ALChar, not a special character
TC-U11	user--name	Invalid	Two consecutive special characters not allowed
TC-U12	user.-name	Invalid	Two different special characters back-to-back not allowed
TC-U13	username-	Invalid	Cannot end on a special character (never returns to accepting state)
TC-U14	"" (empty string)	Invalid	No initial ALChar to consume
TC-U15	user name	Invalid	Space is not ALChar, UNChar, or an allowed special character
TC-U16	user@name	Invalid	@ is not an allowed special character for usernames

2. Password Recognizer

State machine summary: Each input character is classified as upperCase (A–Z), lowerCase (a–z), numericChar (0–9), specialChar (one of ~ ` ! @ # $ % ^ & * ( ) _ - + { } [ ] \ : , . ? /), or otherChar (anything else). charCounter increments per character. On full input consumption: longEnough = True if 8 ≤ charCounter ≤ 32; tooLong = True if charCounter > 32.

ID	Input	Expected Result	Rule Being Tested
TC-P01	Passw0rd! (9 chars)	Valid — upperCase, lowerCase, numericChar, specialChar all True; longEnough True	All four character classes present, valid length
TC-P02	8-character password, e.g. Aa1!Aa1!	longEnough True	Lower boundary of length range
TC-P03	7-character password, e.g. Aa1!Aa1	longEnough False	Just below minimum length
TC-P04	32-character password with mixed classes	longEnough True	Upper boundary of length range

ID	Input	Expected Result	Rule Being Tested
TC-P05	33-character password with mixed classes	tooLong True	Just above maximum recognizer length
TC-P06	"" (empty string)	charCounter = 0; all flags False; longEnough False	No input consumed
TC-P07	alllowercase	lowerCase True only (assuming length valid)	Only-lowercase input
TC-P08	ALLUPPERCASE	upperCase True only	Only-uppercase input
TC-P09	12345678	numericChar True only	Only-digit input
TC-P10	!@#$%^&*	specialChar True only	Only allowed-special-character input
TC-P11	Password! (no digit)	numericChar False; upperCase, lowerCase, specialChar True	Missing one character class (feedback check for US-06)
TC-P12	Password containing é or a control character	otherChar True	Character outside all four defined classes
TC-P13	Password containing ; or '	otherChar True	Character not in the defined special-character list
TC-P14	Valid password with all 4 classes, length exactly 8	All relevant flags True, longEnough True, tooLong False	Combined boundary + completeness case

3. Email Address Validator

State machine summary: Local part: State 0 → State 1 on LPChar; State 1 loops on LPChar; a . moves State 1 → State 0, which requires another LPChar to return to State 1 (so a period can't lead/trail or repeat in the local part). @ moves State 1 → State 2. Domain part: State 2 → State 3 on DPChar; State 3 loops on DPChar; . moves State 3 → State 2 (needs another DPChar to return); - moves State 3 → State 4 (needs another DPChar to return to State 3). EoI (end of input) is only accepted from State 3.

ID	Input	Expected Result	Rule Being Tested
TC-E01	user@example.com	Valid	Basic well-formed address
TC-E02	first.last@example.com	Valid	Single period inside local part
TC-E03	user@my-site.com	Valid	Single dash inside domain
TC-E04	user@mail.example.co.uk	Valid	Multiple domain labels separated by periods
TC-E05	a@b.co	Valid	Minimal-length local part and domain
TC-E06	userexample.com	Invalid	Missing @
TC-E07	.user@example.com	Invalid	Local part cannot start with .
TC-E08	user..name@example.com	Invalid	Consecutive periods not allowed in local part
TC-E09	user.@example.com	Invalid	Local part cannot end on . before @
TC-E10	@example.com	Invalid	Local part cannot be empty (no LPChar before @)
TC-E11	user@	Invalid	Domain part cannot be empty
TC-E12	user@.com	Invalid	Domain cannot start with .
TC-E13	user@example.com.	Invalid	Domain cannot end on . (EoI only valid from State 3)
TC-E14	user@example.com-	Invalid	Domain cannot end on -
TC-E15	user@exa--mple.com	Invalid	Consecutive dashes not allowed in domain
TC-E16	user@example..com	Invalid	Consecutive periods not allowed in domain
TC-E17	us er@example.com	Invalid	Space is not a valid LPChar
TC-E18	user@exam ple.com	Invalid	Space is not a valid DPChar

ID	Input	Expected Result	Rule Being Tested
TC-E19	Big Email	Invalid	Too many characters

4. Functional / User Story Test Cases

These tests validate the Phase 1 user stories themselves (TP1 Plan.pdf / Sprint Backlog.pdf), beyond field-level validation. IDs here are functionality-based (TC-F##) rather than tied to a specific US-number.

4.1 One-time password reset

ID	Scenario	Expected Result	Story Tested
TC-F01	Admin sets a one-time password for a user who forgot their password	User's password is updated to the one-time value and flagged as one-time	Admin can issue a reset
TC-F02	User logs in using a valid one-time password	Login succeeds; user is immediately required to set a new password before reaching the application	Forced password change is enforced
TC-F03	After the new password is set, the same one-time password is used again to log in	Invalid — login rejected	One-time password is cleared after use, cannot be reused
TC-F04	User finishes setting the new password	User is returned to the login screen and must log in again with the new password	Re-login is required after reset

4.2 Delete user account

ID	Scenario	Expected Result	Story Tested
TC-F05	Admin selects a user account and chooses Delete	A confirmation prompt ("Are you sure?") is shown before anything is deleted	Delete requires confirmation
TC-F06	Admin answers "Yes" to the confirmation prompt	Account is removed; that user can no longer log in	Confirmed deletion takes effect
TC-F07	Admin declines / closes the confirmation prompt	Account is not deleted	Cancelling the confirmation preserves the account
TC-F08	Admin attempts to delete their own currently logged-in account	Invalid — action blocked	Self-deletion is prevented

4.3 List all user accounts

ID	Scenario	Expected Result	Story Tested
TC-F09	Admin opens the account listing screen	Every account is listed with username, full name, email address, and assigned role(s)	Admin can list all accounts
TC-F10	A new account is created, then the listing is refreshed	The new account appears in the list with correct details	Listing reflects current account data

4.4 Add / remove roles per user

ID	Scenario	Expected Result	Story Tested
TC-F11	Admin adds a role to an existing user	The role appears in that user's assigned roles and the user gains the associated access	Admin can add a role

ID	Scenario	Expected Result	Story Tested
TC-F12	Admin removes a role from an existing user	The role no longer appears for that user and associated access is revoked	Admin can remove a role
TC-F13	Admin attempts to remove a user's only remaining role	Team decision needed: whether a user must always retain at least one role	Open item confirm expected behavior with the team before finalizing this test

4.5 Replace placeholder roles (Role1 / Role2)

ID	Scenario	Expected Result	Story Tested
TC-F14	Inspect every screen and code path where Role1 / Role2 previously appeared	No placeholder role names remain visible anywhere in the running application	Placeholder roles fully replaced
TC-F15	Assign one of the new real roles to a user	Role behaves consistently with the existing role-handling logic (login, access, listing)	New roles follow the same pattern as prior roles